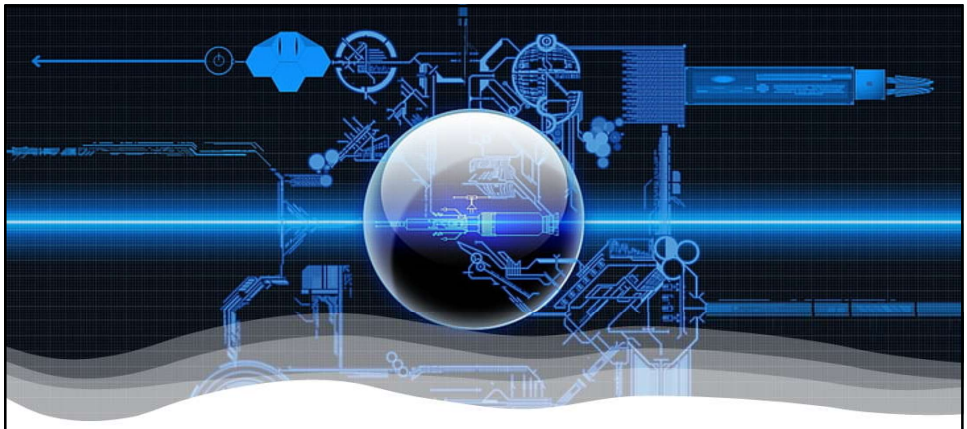


CS 442 (F2025)
Mobile Application Development

Global System for Mobile Communications (GSM)

1

https://www.gsma.com/

- More than 70% of all the digital wireless market
- Business objectives (https://www.gsmaintelligence.com/)

  - Support for international roaming
  - Good speech quality
  - Ability to support handheld terminals
  - Low terminal and service cost
  - Spectral efficiency
  - Support for a range of new services and facilities
  - ISDN compatibility.

2

GSM History

https://www.gsma.com/newsroom/press-release/gsma-highlights-pakistans-digital-progress-and-potential-to-spur-accelerated-economic-growth/

Year	Event
1982	Groupe Spécial Mobile (GSM) established
1987	Essential elements of wireless transmission specified
1989	GSM become an ETSI technical committee
1990	Phase 1 GSM 900 specification (designed 1987 through 1990) frozen
1991	First GSM network launched
1993	First roaming agreement came into effect
1994	Data transmission capability launched
1995	Phase 2 launched. Fax and SMS roaming services offered
2002	SMS volume crosses 24 billions/year, 750 millions subscribers

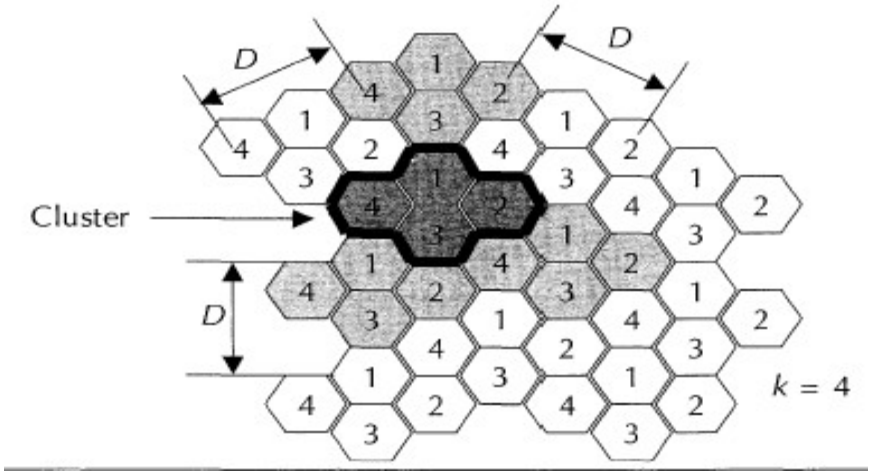
3

Frequency Reuse - Characteristics

- The area to be covered is subdivided into radio zones or cells (Figure 5.1). Though in reality these cells could be of any shape, for convenient modeling purposes these are modeled as hexagons. Base stations are positioned at the center of these cells.
- Each cell i receives a subset of frequencies fb_i from the total set assigned to the respective mobile network. To avoid any type of co-channel interference, two neighboring cells never use the same frequencies.
- Only at a distance of D (known as frequency reuse distance), the same frequency from the set fb_i can be reused. Cells with distance D from cell i, can be assigned one or all the frequencies from the set fb_i belonging to cell i.
- When moving from one cell to another during an ongoing conversation, an automatic channel change occurs. This phenomenon is called handover. Handover maintains an active speech and data connection over cell boundaries.

4

Cell clusters in GSM



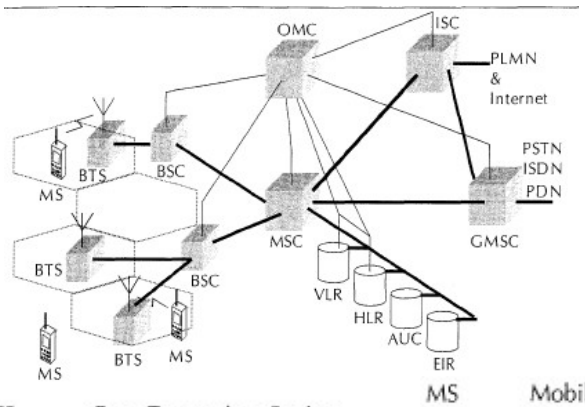
5

Frequency Reuse

The regular repetition of frequencies in cells result in a clustering of cells. The clusters generated in this way can consume the whole frequency band. The size of a cluster is defined by *k*, the number of cells in the cluster. This also defines the frequency reuse distance *D*. Figure 5.1 shows an example of cluster size of 4.

6

GSM Architecture



BTS	Base Transceiver Station
BSC	Base Station Controller
MSC	Mobile Switching Center
GMSC	Gateway MSC
ISC	International Switching Center
MS	Mobile Station
HLR	Home Location Register
VLR	Visitor Location Register
AUC	Authentication Center
EIR	Equipment Identity Register
OMC	Operation and Maintenance Center

7

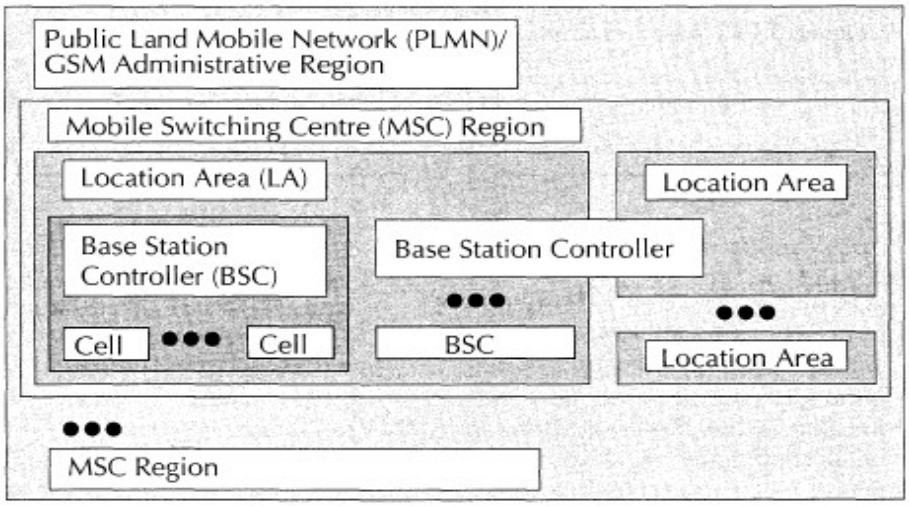
GSM Network

GSM networks are structured in hierarchic fashion (Figure 5.2). It consists at the minimum one administrative region assigned to one MSC (Mobile Switching Centre). The administrative region is commonly known as PLMN (Public Land Mobile Network). Each administrative region is subdivided into one or many Location Area (LA). One LA consists of many cell groups. Each cell group is assigned to one BSC (Base Station Controller). For each LA there will be at least one BSC. Cells in one BSC can belong to different LAs.

Cells are formed by the radio areas covered by a BTS (Base Transceiver Station) (Figure 5.3). Several BTSs are controlled by one BSC. Traffic from the MS (Mobile Station) is routed through MSC. Calls originating from or terminating in a fixed network or other mobile networks is handled by the GMSC (Gateway MSC). Figure 5.3 depicts the architecture of a GSM PLMN from technology point of view, whereas Figure 5.4 depicts the same architecture from operational point of view.

8

GSM System Hierarchy



9

Permanent Data

For all subscribers registered with a cellular network operator, permanent data such as the service profile is stored in the Home Location Register (HLR). The data relate to the following information:

- Authentication information like International Mobile Subscriber Identity (IMSI)
- Identification information like name, address, etc. of the subscriber
- Identification information like Mobile Subscriber ISDN (MSISDN) etc.
- Billing information like prepaid or postpaid
- Operator selected denial of service to a subscriber
- Handling of supplementary services like for CFU (Call Forwarding Unconditional), CFB (Call Forwarding Busy), CFNR (Call Forwarding Not Reachable) or CFNA (Call Forwarding Not Answered)
- Storage of SMS Service Center (SC) number in case the mobile is not connectable so that whenever the mobile is connectable, a paging signal is sent to the SC

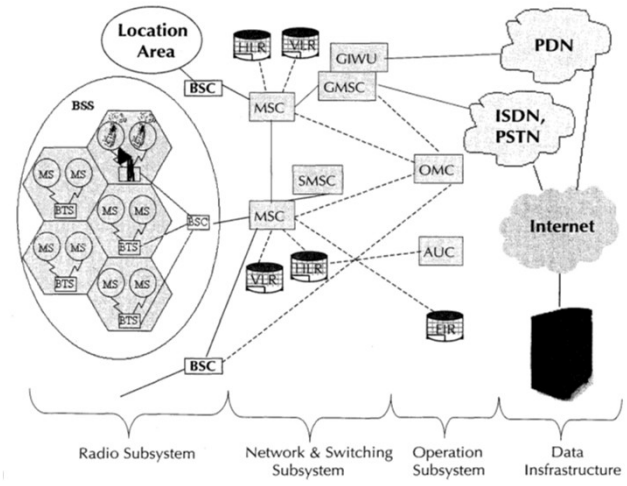
10

- Provisioning information like whether long distance and international calls allowed or not
- Provisioning information like whether roaming is enabled or not
- Information related to auxiliary services like Voice mail, data, fax services etc.
- Information related to auxiliary services like CLI (Caller Line Identification) etc.
- Information related to supplementary services for call routing. In GSM network one can customize the personal profile to the extent that while the subscriber is roaming in a foreign PLMN, incoming calls can be barred. Also, outgoing international calls can be barred etc.

There is some variable information, which could also be part of the HLR. This includes the pointer to the VLR, location area of the subscriber, Power OFF status of the handset etc.

11

System Architecture of GSM



12

BTS	Base Transceiver Station	ISDN	Integrated System Digital Network
BSC	Base Station Controller	HLR	Home Location Register
MSC	Mobile Switching Center	VLR	Visitor Location Register
GMSC	Gateway MSC	AUC	Authentication Center
SC	Service Center/SMS Controller	EIR	Equipment Identity Register
GIWU	Gateway Interworking Unit	OMC	Operation and Maintenance Center
PDN	Public Data Network	PSTN	Public Switched Telephone Network
MS	Mobile Station		

13

GSM Entities

The GSM technical specifications define different entities that form the GSM network by defining their functions and interface requirements. The GSM network can be divided into four main groups (Figure 5.4):

- The Mobile Station (MS). This includes the Mobile Equipment (ME) and the Subscriber Identity Module (SIM).
- The Base Station Subsystem (BSS). This includes the Base Transceiver Station (BTS) and the Base Station Controller (BSC).
- The Network and Switching Subsystem (NSS). This includes Mobile Switching Center (MSC), Home Location Register (HLR), Visitor Location Register (VLR), Equipment Identity Register (EIR), and the Authentication Center (AUC).
- The Operation and Support Subsystem (OSS). This includes the Operation and Maintenance Center (OMC).

14

Mobile Station

Mobile Station is the technical name of the mobile or the cellular phone. In early days mobile phones were a little bulky and were sometimes installed in cars like other equipments. Even the handheld terminals were quite big. Though the phones have become smaller and lighter, they are still called Mobile Stations. MS consists of two main elements:

- The mobile equipment or the mobile device. In other words, this is the phone without the SIM card.
- The Subscriber Identity Module (SIM).

There are different types of terminals distinguished principally by their power and application. The handheld GSM terminals have experienced the highest evolution. The weight and volume of these terminals are continuously decreasing. The life of a battery between charging is also increasing. The evolution of technologies allowed decrease of this power to 0.8 W.

15

The SIM is installed in every GSM phone and identifies the terminal. Without the SIM card, the terminal is not operational. The SIM cards used in GSM phones are smart processor cards. These cards posses a processor and a small memory. By inserting the SIM card into the terminal, the user can have access to all the subscribed services. The SIM card contains the International Mobile Subscriber Identity (IMSI) used to identify the subscriber to the system, a secret key for authentication, and other security information. Another advantage of the SIM card is the mobility of the users. In fact, the only element that personalizes a terminal is the SIM card. Therefore, the user can have access to its subscribed services in any terminal using his or her SIM card.

16

Base Station Subsystem

The BSS (Base Station Subsystem) connects the Mobile Station and the NSS (Network and Switching Subsystem). It is in charge of the transmission and reception for the last mile. The BSS can be divided into two parts:

- The Base Transceiver Station (BTS) or Base Station in short.
- The Base Station Controller (BSC).

The Base Transceiver Station corresponds to the transceivers and antennas used in each cell of the network. In a large urban area, a large number of BTSs are potentially deployed. A BTS is usually placed in the center of a cell. Its transmitting power defines the size of a cell. The BTS houses the radio transmitter and the receivers that define a cell and handles the radio-link protocols with the Mobile Station. Each BTS has between one and sixteen transceivers depending on the density of users in the cell.

Base Station Controller is the connection between the BTS and the Mobile service Switching Center (MSC). The BSC manages the radio resources for one or more BTSs. It handles handovers, radio-channel setup, control of radio frequency power levels of the BTSs, exchange function, and the frequency hopping.

17

Network and Switching Subsystem

The central component of the Network Subsystem is the Mobile Switching Center (MSC). It does multiple functions:

- It acts like a normal switching node for mobile subscribers of the same network (connection between mobile phone to mobile phone within the same network).
- It acts like a normal switching node for the PSTN fixed telephone (connection between mobile phone to fixed phone).
- It acts like a normal switching node for ISDN.
- It provides all the functionality needed to handle a mobile subscriber, such as registration, authentication, location updating, handovers and call routing.
- It includes databases needed in order to store information to manage the mobility of a roaming subscriber.

18

These different services are provided in conjunction with several functional entities, which together form the Network Subsystem. The signaling between functional entities in the Network Subsystem uses Signaling System Number 7 (SS7). SS7 is used for trunk signaling in ISDN and widely used in today's public networks. SS7 is also used for SMS, prepaid, roaming and other intelligent network functions.

The MSC together with Home Location Register (HLR) and Visitor Location Register (VLR) databases, provide the call-routing and roaming capabilities of GSM. The HLR is considered a very important database that stores information of subscribers belonging to the covering area of a MSC. Although a HLR may be implemented as a distributed database, there is logically only one HLR per GSM network. The HLR contains all the administrative information of each subscriber registered in the corresponding GSM network. This includes information like current location of the mobile, all the service provisioning information and authentication data. When a phone is powered off, this information is stored in the HLR. The location of the mobile is typically in the form of the signaling address of the VLR associated with the mobile station. HLR is always fixed and stored in the home network, whereas the VLR logically moves with the subscriber.

19

The VLR can be considered a temporary copy of some of the important information stored in the HLR. VLR is similar to a cache, whereas HLR is the persistent storage. The VLR contains selected administrative information borrowed from the HLR, necessary for call control and provisioning of the subscribed services. This is true for each mobile currently located in the geographical area controlled by a VLR. GSM standards define interfaces to HLR; however, there is no interface standard for VLR. Although

Within the NSS there is a component called Gateway MSC (GMSC) that is associated with the MSC. A gateway is a node interconnecting two networks. The GMSC is the interface between the mobile cellular network and the PSTN. It is in charge of routing calls from the fixed network towards a GSM user and vice versa. The GMSC is often implemented in the same node as the MSC. Like the GMSC, there is another node called GIWU (GSM Interworking Unit). The GIWU corresponds to an interface to various networks for data communications. During these communications, the transmission of speech and data can be alternated.

20

Operations and Support System

As the name suggests, Operations and Support Subsystem (OSS) controls and monitors the GSM system. The OSS is connected to the different components of the NSS and to the BSC. It is also in charge of controlling the traffic load of the BSS. However, the increasing number of base stations, due to the development of cellular radio networks, has resulted in some of the maintenance tasks being transferred to the BTS. This transfer decreases considerably the costs of the maintenance of the system. Provisioning information for different services is managed in this layer.

Equipment Identity Register (EIR) is a database that contains a list of all valid mobile equipment within the network, where each mobile station is identified by its International Mobile Equipment Identity (IMEI). EIR contains a list of IMEIs of all valid terminals. An IMEI is marked as invalid if it has been reported stolen or is not type approved. The EIR allows the MSC to forbid calls from this stolen or unauthorized terminals.

Authentication Center (AUC) is responsible for the authentication of a subscriber. This is a protected database and stores a copy of the secret key stored in each subscriber's SIM card. These data help to verify the user's identity.

21

Short Message Service (SMS)

Short message or SMS is one of the most popular services within GSM. SMS is a data service and allows a user to enter text message up to 160 characters in length when 7-bit English characters are used. It is 140 octets when 8-bit characters (some European alphabets or binary data) are used, and 70 characters in length when non-Latin alphabets such as Arabic, Chinese or Hindi are used (70 characters of 16-bit Unicode). SMS is a proactive bearer and is an **always ON** network. Message center is also referred to as Service Centre (SC) or SMS Controller (SMSC). SMSC is a system within the core GSM network, which works as a the store and forward system for SMS messages. Refer to Figure 5.5 for SMS architecture.

There are two types of SMS, SMMT (Short Message Mobile Terminated Point-to-Point), and SMMO (Short Message Mobile Originated Point-to-Point). SMMT is an incoming short message from the network and is terminated in the MS (phone or Mobile Station). SMMO is an outgoing message, originated in the MS, and forwarded to the network for delivery. For an outgoing message, the SMS is sent from the phone to SC via the VLR and the Interworking MSC (IWMSC). For incoming SMS message the path is from SC to the MS via the HLR and the Gateway MSC (GMSC). Please see chapter

22

- GSM supports data access over CSD (Circuit Switch Data)
- CSD uses a single radio time slot to deliver 9.6 kbit/s data transmission to the GSM Network and Switching Subsystem where it could be connected through the equivalent of a normal modem to the Public Switched Telephone Network (PSTN) allowing direct calls to any dial-up service.
- WAP is the data service supported by GPRS (General Packet Radio Service) and GSM to access Internet and remote data services

23

History

- The SMS concept was developed in the Franco-German GSM cooperation in 1984 by Friedhelm Hillebrand and Bernard Ghillebaert.
- 1992: first SMS. SEMA, British Technology Firm
- 1999: SMS got popular
- Today SMS is bearer/service with in GSM
- 2.4 billion active users, or 74% of all mobile phone subscribers

24

. The SS#7 signaling channels are always physically present but mostly unused, be it during an active user connection or in the idle state. It is, therefore, quite an attractive proposition to use these channels for transmission of used data. SMS uses the free capacity of the signaling channel. Each short message is up to 160 characters in length when 7-bit English characters are used. It is 140 octets when 8-bit characters (some European alphabets) are used, and 70 characters in length when non-Latin alphabets such as Arabic, Chinese or Hindi are used (70 characters of 16 bit Unicode).

25

Message

- Transmission of short messages between the SMSC and the handset is done whenever using the Mobile Application Part (MAP) of the SS7 protocol. Messages are sent with the MAP MO- and MT-ForwardSM operations, whose payload length is limited by the constraints of the signaling protocol

26

Message Size

- Precisely 140 octets (140 octets = 140 * 8 bits = 1120 bits). Short messages can be encoded using a variety of alphabets: the default GSM 7-bit alphabet, the 8-bit data alphabet, and the 16-bit UTF-16 alphabet.
- Larger content (concatenated SMS, multipart or segmented SMS, or "long SMS") can be sent using multiple messages, in which case each message will start with a user data header (UDH) containing segmentation information. Since UDH is part of the payload, the number of available characters per segment is lower: 153 for 7-bit encoding, 134 for 8-bit encoding and 67 for 16-bit encoding.

27

Strengths of SMS

Omnibus nature of SMS: SMS uses SS7 signaling channel, which is available throughout the world. SMS is the only bearer that allows a subscriber to send a long distance SMS without having long distance subscription. For example, you cannot make a voice call to a mobile phone in UK unless you have an international calling facility. However, you can send a SMS to a subscriber in UK, without having an international call facility.

Stateless: SMS is sessionless and stateless. Every SMS message is unidirectional and independent of any context. This makes SMS the best bearer for notifications, alerts and paging. SMS can be used for proactive information dissemination for 'unsolicited response' and business triggers generated by applications (referred as 'Push' Figure 8.4).

28

Asynchronous: In http, for every command (e.g., GET or POST) there is a request and a response pair making it synchronous at the transaction level. Unlike http, SMS is completely asynchronous. In case of SMS, even if the recipient is out of service, the transmission will not be abandoned. Therefore, SMS can be used as message queues. In essence, SMS can be used as a transport bearer for both synchronous (transaction oriented) and asynchronous (message queue and notification) information exchange.

Self-configurable and last mile problem resistant: SMS is self-configurable. In the case of Web or WAP, it is no trivial task to connect to a service from a foreign network without any change in the configuration or preference setting. The device needs to be configured interactively by the user or system administrator to access the network. This makes the access dependent on the last mile. SMS has no such constraints. While in a foreign network, one can access the SMS bearer without any change in the phone settings. The subscriber is always connected to the SMS bearer irrespective of the home and visiting network configurations. While roaming in a foreign network, even if the serving network does not have a SMSC (SMS Center) or SC (Service Center), SMS can be sent and received.

29

Non-repudiable: SMS message carries the SC and the source MSISDN as a part of the message header. Unlike an IP address it is not easy to handcraft a MSISDN address in the SMS. It is possible for an application connected to an SMS to handcraft a MSISDN address like “999” or even alphabetic addresses like "MYBANK". However, an application can not handcraft the SC address. Therefore, an SMS can prove beyond doubt the origin of the SMS.

Always connected: As SMS uses the SS7 signaling channel for its data traffic, the bearer media is always on. User cannot switch OFF, BAR or DIVERT any SMS message. When a phone is busy and a voice, data or FAX call is in progress, SMS message is delivered to the MS (Mobile Station) without any interruption to the call.

30

SMS types

SMS are basically of two types, **SM MT** (Short Message Mobile Terminated Point-to-Point), and **SM MO** (Short Message Mobile Originated Point-to-Point). SM MT is an incoming short message from the network side and is terminated in the MS. SM MO is an outgoing message, originated in the user device (MS), and forwarded to the network

31

SMS as an Information Bearer

To use SMS as a bearer for any information service, we need to connect the services running on the Enterprise Origin server to the SC through an SME (Short Message Entity) or ESME (External Short Message Entity). SME in any network is generally a SMS gateway. With respect to SMS, a GSM subscriber is always in control of the SC in the home network irrespective of the serving network. Thus, if there is any SMS-based data service in the home network, it will be available in any foreign network.

32

Operator Centric

For a SM MO to work it is mandatory that a SC is used. As a part of SMS value added services, operators offer different information on demand and entertainment services. These are done through connecting an Origin server to the SC via a SMS gateway. In different parts of the world a new industry vertical has emerged to address this market. These service providers are known as MVNO (Mobile Virtual Network Operators).

33

Operator Independent Push

We have seen that it is possible to send a SMS to any phone in any network. For example, a MT message can be delivered from a network in India to a MS of UK roaming in Germany (Figure 6.5). Which in other words means that any push, which may be an alert, notification or even response from a pull message generated by an application, can be serviced by any network and delivered to any GSM phone in any network without any difficulty.

Assuming that appropriate roaming tie-ups are in place, an enterprise can use SMS to send business alerts or proactive notifications to its customer anywhere, anytime on any GSM phone. With roaming tie-ups, operators reach an agreement on revenue share and call forwarding mechanism. Roaming tie-ups are a commercial issue rather than technical. Some credit card companies in India send SMS notifications to its cardholders in different networks using operator independent push.

34

Operator Independent Pull

As the SME is always connected to the home network's SC, with the conventional framework, it is not possible to route mobile originated SMS messages to any application or any SME of choice. There are ways by which a SMS message can be routed to some enterprise SME connected to external SC. This is achieved through SAT, where the SAT application running on the SIM card changes the SC number during the transmission of the SMS and forces the SMS to recognize a different SC of a different network as its home SC. In this case also, technically the SMS is sent to the SME connected to the home SC. SMS has always been considered a revenue generating tool for cellular operators. Therefore, the current framework suits a cellular operator very well. If a SMS service is operator dependent, the cellular operator can use this to its advantage. In today's global scenario an enterprise or a MVNO has its customers around the world subscribing to different GSM networks. To make this possible, enterprises need operator-independent pull as well. Operator-independent pull services can be achieved using GSM modem technology described in the following sections. Also, the same can be done using Intelligent Network Technologies.

35

Value added services

News/Stock Quotes Service

In a service like **News** or **Stock Quote**, we get the latest news or stock information. This will be a Short transaction. The keyword for news will be **News**, whereas the keyword for stock quote can be **BSE**. BSE Infosys, will give the stock price of Infosys at the Bombay Stock Exchange. These are examples for real-time information on demand. For services like **News** and **Stock Quote** we need to have a relationship with some content provider who will supply us the up-to-date information. For example, we could tie up with CNN for international news, Indian Express for general news, weather.com for weather news etc. For stock quote, we may need to tie up with a stock exchange like Bombay Stock Exchange or National Stock Exchange. We will receive live feed from these content providers and update the content database on a real-time basis. As and when a subscriber wants these information, we supply the latest information from the live database.

36

Session-based Chat Application

A chat service is essentially a session oriented transaction. In a chat service the user needs to log in. The user needs to explicitly log out or will be logged out implicitly following a period of inactivity. Every time the user sends a chat keyword, we need to know the previous transactions. Every SMS message carries the unique MSISDN number.

This unique MSISDN number of the phone can be used as the session key. In the chat software we remember the state of the transaction using this MSISDN.

37

Email through SMS

This is a very useful service and is a transaction-oriented dialogue. To send an email through SMS, the user message will be **mail roopa@iitb.ac.in we will meet tomorrow 6:00 pm**. This VAS will send a mail to Roopa with mail id roopa@iitb.ac.in. The body of the mail will be 'we will meet tomorrow 6:00 pm.' The mail will be sent to Roopa by a SMTP server.

Health Care Services

Health care applications need both Pull and Push. A typical health care application could be ICU (Intensive Care Unit) system. The system will include alerts to doctor. In status monitoring service, a doctor or a nurse can enquire the status of a patient in the ICU. A limited enquiry facility will be provided to one MSISDN outside the hospital staff. This could be for someone in the family. This enquiry will be a Short transaction. We will have alert services as well. In the alert service, nurses and doctors are notified periodically about the status of the patients in the ICU. The alerts can also be integrated with medical equipment.

38

Micro-Payment Services

Let us take an example of micro-payment for a vending machine. This will be a session-oriented dialogue. In this application there will be some id pasted on the vending machine. The user enters this number and sends to the VAS. The VAS will authenticate the user and check whether the user has sufficient money in credit. Based upon the credit the transaction will either be approved or rejected. If approved, an authorization message will be sent back to the vending machine. The vending machine will ask the user to select merchandise. The user selects the merchandise, a soft drink for example. Once the merchandise is dispensed, the vending machine will send back a message to the VAS indicating that the merchandise has been dispensed. The price for the merchandise is debited from the user account.

39

Alert Services

These are proactive alert services. For a stock quote the alert services can be of the following kind.

Time-based: In this service, proactive alerts are sent to the mobile phone at a pre-assigned time of the day. The alert contains the stock quote of different scripts of the portfolio.

Watermark based: In this service whenever the stock price goes up or falls down to a certain level, alerts are sent. This information will help the subscriber to decide whether to buy or sell some particular stock.

For other services, like cricket score, it can be a periodic alert (every 10 minutes) during the match. There can be other alerts like inform the live score whenever a player is being out etc.

40

Location Based Software

Device-specific location awareness requires either of the following technologies:

1. Cell ID (CID)-based system.
2. Global Positioning System (GPS)-based system.

Location-based services could be road direction, restaurant guide etc. Some location-aware VAS services provide shopping alerts as well. In location-based services only the information relevant to the current location of the mobile phone (or the subscriber) is provided. In a shopping service, the user will receive alerts on discount or sale information when they pass through close the proximity of the shopping malls. In the case of a restaurant guide, let us assume that the subscriber is in a office on M.G. Road in Bangalore and sends **Res** to the VAS. Only the restaurants in, and around, M.G. Road will be provided as response to this request. When the same user asks for the same information in Mumbai, restaurants

41

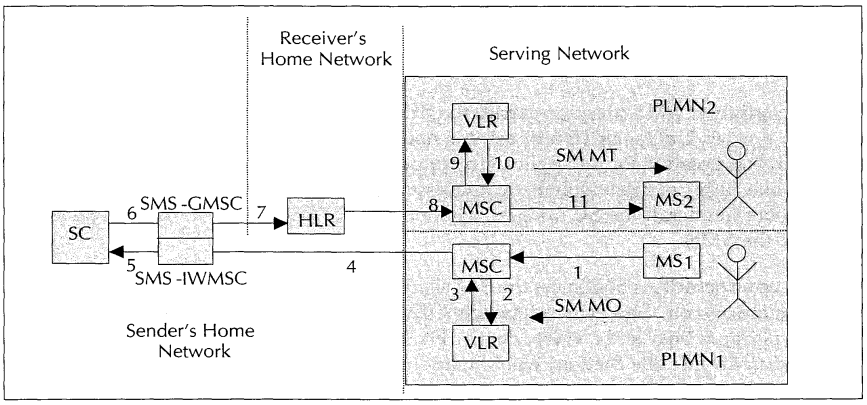


Figure 6.1 The main network structure serving as a basis for the short message transfer

42

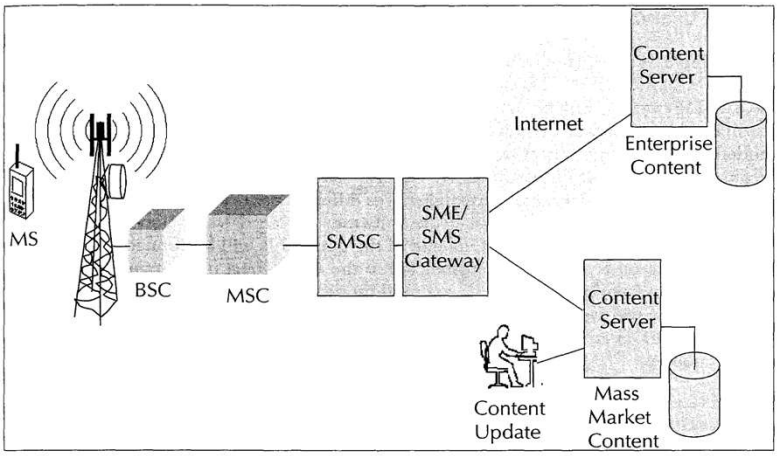


Figure 6.6 SMS Value Added Services Architecture

43